

3 common cloud security threats with real-world examples

1. Cloud Misconfigurations

- **Threat Overview:**

Misconfigurations happen when cloud services are not set up securely—like leaving storage buckets publicly accessible or not properly restricting user permissions.

- **Real-World Example:**

Capital One Data Breach (2019)

A misconfigured AWS S3 bucket allowed a former AWS employee to access over 100 million Capital One customer records. The attacker exploited a Server-Side Request Forgery (SSRF) vulnerability.

- **Impact:**

1. 100M+ customer records breached
 2. ~\$80M in fines and remediation costs
 3. Huge reputational damage
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2. Insecure APIs

- **Threat Overview:**

Cloud services rely heavily on APIs for functionality. If these APIs are insecure—lack proper authentication or rate limiting—they become a gateway for attackers.

- **Real-World Example:**

Facebook (Meta) API Misuse

Facebook APIs allowed apps like Cambridge Analytica to access data from millions of users without their direct consent, sparking a massive privacy scandal.

- **Impact:**

1. Data misuse of up to 87M users
 2. Congressional hearings and regulatory scrutiny
 3. \$5B fine by the FTC
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3. Data Leaks and Exposure

- **Threat Overview:**

Data stored in the cloud is vulnerable to leakage if not encrypted or access-controlled properly. This often happens due to negligence or lack of visibility over resources.

- **Real-World Example:**

Accenture (2017)

Cybersecurity researchers found Accenture's AWS S3 buckets were publicly accessible, exposing sensitive data such as API data, credentials, and decryption keys.

- **Impact:**

1. Credentials and secret keys exposed
2. Potential access to internal systems
3. Risk of client data being accessed